Week 8 MAT110

DIFFERENTIAL CALCULUS & COORDINAE GEOMETRY

TAYLOR Expansion of a two VARIABLE FUNCTION

Taylor expansion of single variable function:

$$f_n(x) = f(a) + (x-a)f'(a) + \frac{(x-a)^2}{2!}f''(a) + \frac{(x-a)^3}{3!}f'''(a) + \dots + \frac{(x-a)^n}{n!}f^{(n)}(a).$$

Let $w = f(x,y)$ is a function of two variable.

The Taylor series expansion for this function with respect to a pt (a,b) is defined as:

$$f(x,y) = \sum_{n=0}^{\infty}\sum_{m=0}^{\infty} C_{n,m}\,(x-a)^n(y-b)^m$$

where $$C_{n,m} = \frac{1}{n!\,m!}\frac{\partial^{(n+m)}f}{\partial x^n \partial y^m}\Big|_{(a,b)}.$$

Expand:

$$f(x,y) = C_{0,0} + C_{1,0}(x-a) + C_{0,1}(y-b) + C_{2,0}(x-a)^2 + C_{1,1}(x-a)(y-b) + C_{0,2}(y-b)^2 + C_{3,0}(x-a)^3 + C_{2,1}(x-a)^2(y-b) + C_{1,2}(x-a)(y-b)^2 + C_{0,3}(y-b)^3 + \dots$$

$C_{0,0} = f(x,y)$

$C_{1,0} = \frac{\partial f}{\partial x}$

$C_{0,1} = \frac{\partial f}{\partial y}$

$C_{1,1} = \frac{\partial^2 f}{\partial x \partial f}$

$C_{2,0} = \frac{1}{2!}\frac{\partial^2 f}{\partial x^2}$

$C_{n,0} = \frac{1}{n!}\frac{\partial^{(n)} f}{\partial x^{(n)}}$

$C_{0,n} = \frac{1}{n!}\frac{\partial^{(n)} f}{\partial y^{(n)}}$

$C_{n,m} = \frac{1}{n!}\frac{1}{m!}\frac{\partial^{(n+m)} f}{\partial x^{(n)}\partial y^{(n)}}$

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Example Compute the 1st and 2nd Degree Taylor Polynomial for $f(x,y) = xe^y + 1$ for (x,y) near the pt $(1,0)$.

Note 1st Degree $= L(x,y)$

2nd Degree $= Q(x,y)$

$$f(x,y) = xe^y + 1$$

$$f_x = \frac{\partial f}{\partial x} = e^y \qquad f_y = \frac{\partial f}{\partial y} = xe^y$$

$$f_{xx} = \frac{\partial^2 f}{\partial x \partial x} = \frac{\partial^2 f}{\partial x^2} = 0 \qquad f_{yy} = \frac{\partial^2 f}{\partial y^2} = xe^y$$

$$\because \frac{\partial f}{\partial x} = e^y$$

$$f_{xy} = \frac{\partial f}{\partial x \partial y} = e^y$$

$$f_{yx} = \frac{\partial f}{\partial y \partial x} = e^y$$

$f_x = e^y$ ——— $f_x(1,0) = e^0 = 1$

$f_y = xe^y$ ——— $f_y(1,0) = 1 \cdot e^0 = 1$

$f_{xx} = 0$ ——— $f_{xx}(1,0) = 0$

$f_{yy} = xe^y$ ——— $f_{yy}(1,0) = 1e^0 = 1$

$f_{xy} = e^y$ ——— $f_{xy}(1,0) = e^0 =$

$f(x,y) = xe^y + 1$ ——— $f(1,0) = 1e^0 + 1 = 2$

1st degree

$$L(x,y) = f(1,0) + f_x(1,0)(x-1) + f_y(1,0)(y-0)$$

$$= 2 + 1(x-1) + 1(y)$$

$$= 2 + x - 1 + y$$

$$= 1 + x + y$$

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2nd degree:

$$Q(x,y) = L(x,y) + \frac{f_{xx}(1,0)}{2!}(x-1)^2 + \frac{f_{xy}(1,0)}{1!\,1!}(x-1)(y-0)$$

$$+ \frac{f_{yy}(1,0)}{2!}\cdot(y-0)^2$$

$$= (1+x+y) + \frac{0}{2!}(x-1)^2 + 1(x-1)(y-0) + \frac{1}{2!}(y-0)^2$$

$$= 1 + x + y + 0 + xy - y + \frac{1}{2}y^2$$

$$= 1 + x + xy + \frac{1}{2}y^2$$

<u>Expand</u>: (Recall)

$$f(x,y) = f(a,b) + \frac{f_x(a,b)}{1!}(x-a) + \frac{f_y(a,b)}{1!}(y-b)$$

$$+ \frac{f_{xx}(a,b)}{2!}(x-a)^2 + \frac{f_{xy}(a,b)}{1!\,1!}(x-a)(y-b) + \frac{f_{yy}(a,b)}{2!}(y-b)^2$$

$$+ \frac{f_{xxx}(a,b)}{3!}(x-a)^3 + \frac{f_{xxy}(a,b)}{2!\,1!}(x-a)^2(y-b) + \frac{f_{xyy}(a,b)}{1!\,2!}(x-a)(y-b)^2$$

$$+ \frac{f_{yyy}(a,b)}{3!}(y-b)^3 + \cdots$$

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